

Question 1

P3

There are 24 different movies Kiandra can download to her computer. Determine the number of ways she can select 15 movies.

Solution

$${}_{24}C_{15} = 1\,307\,504 \text{ combinations}$$

1 mark

Given $\theta = 40^\circ$,

- a) convert θ to radians.
- b) determine the coterminal angles of θ where $\theta \in \mathbb{R}$.

Solution

a)

$$\theta = 40 \left(\frac{\pi}{180} \right)$$

$$\theta = \frac{2\pi}{9}$$

or

$$\theta = 0.698$$

1 mark

b)

$$\theta = \frac{2\pi}{9} + 2\pi k \in \mathbb{Z}$$

or

$$\theta = 0.698 + 2\pi k \in \mathbb{Z}$$

or

$$\theta = 40^\circ + 360^\circ k, k \in \mathbb{Z}$$

1 mark

Question 3

R10

Peter invests \$560 per month at an annual interest rate of 4.2%, compounded monthly. Determine how many monthly investments he will need to make to obtain at least \$500 000. Express your answer as a whole number.

Use the formula:

$$FV = \frac{R \left[(1+i)^n - 1 \right]}{i}$$

where FV = the future value

R = the investment amount each period

$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$

n = the number of investments

Solution

$$500\,000 = \frac{560 \left[\left(1 + \frac{0.042}{12} \right)^n - 1 \right]}{\frac{0.042}{12}}$$

½ mark for substitution

$$500\,000 = \frac{560 \left[(1 + 0.0035)^n - 1 \right]}{0.0035}$$

$$500\,000 = 160\,000 (1.0035^n - 1)$$

$$3.125 = 1.0035^n - 1$$

$$4.125 = 1.0035^n$$

½ mark for simplification

$$\log 4.125 = \log 1.0035^n$$

½ mark for applying logarithms

$$\log 4.125 = n \log 1.0035$$

1 mark for power law

$$n = \frac{\log 4.125}{\log 1.0035}$$

$$n = 405.584$$

½ mark for solving for n

∴ 406 monthly investments are needed.

3 marks

Question 4

P2

Ishmael has 4 dogs, 5 cats, and 3 horses.

If he arranges all of them in a row, determine how many ways they can be arranged if each type of animal must be grouped together.

Solution

$$\frac{3!}{\text{types of animals}} \cdot \frac{4!}{\text{dogs}} \cdot \frac{5!}{\text{cats}} \cdot \frac{3!}{\text{horses}} = 103\,680 \text{ ways}$$

1 mark for arrangement of types of animals
1 mark for arrangement of dogs, cats, and horses

2 marks

Solve the following equation algebraically over the interval $0 \leq \theta \leq 2\pi$.

$$2 \cos^2 \theta + 9 \cos \theta - 5 = 0$$

Solution

$$2 \cos^2 \theta + 9 \cos \theta - 5 = 0$$

$$(2 \cos \theta - 1)(\cos \theta + 5) = 0$$

$$\cos \theta = \frac{1}{2}$$

$$\cos \theta = -5$$

1 mark for solving for $\cos \theta$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

no solution

2 marks for solving for θ

($\frac{1}{2}$ mark for each value, 1 mark for indicating no solution)

3 marks

Determine which term contains $\frac{1}{x^6}$ in the binomial expansion of $\left(\frac{2}{x^3} + 3x^2\right)^7$.

Solution

Method 1

$$\left(x^{-3}\right)^{7-k} \left(x^2\right)^k = x^{-6}$$

½ mark for substitution

$$x^{-21+3k} x^{2k} = x^{-6}$$

$$x^{-21+5k} = x^{-6}$$

$$-21 + 5k = -6$$

$$5k = 15$$

$$k = 3$$

½ mark for solving for k

∴ the fourth term contains $\frac{1}{x^6}$

1 mark for the 4th term (or a term consistent with the value of k)

2 marks

Method 2

$$\left(\frac{1}{x^3}\right)^7, \left(\frac{1}{x^3}\right)^6 (x^2), \left(\frac{1}{x^3}\right)^5 (x^2)^2, \dots$$

$$\frac{1}{x^{21}}, \frac{1}{x^{16}}, \frac{1}{x^{11}}, \dots$$

1 mark for determining a pattern

∴ the fourth term contains $\frac{1}{x^6}$

1 mark for the 4th term (or a term consistent with the pattern)

2 marks

Question 7

T1

Determine the radius of a circle which has an arc length of 5 cm with a central angle of 3 radians.

Solution

$$s = \theta r$$

$$5 = 3r$$

$$r = \frac{5}{3} \text{ cm}$$

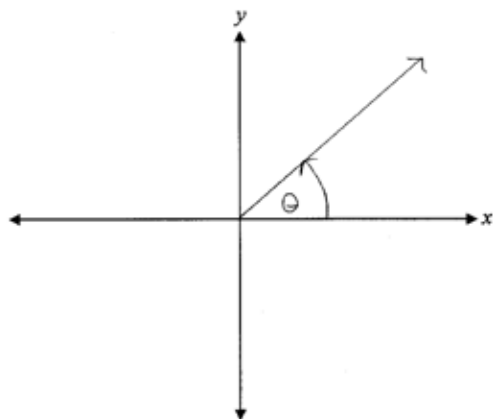
1 mark

Question 8

T1

Tyler incorrectly sketched the angle $\theta = -\frac{7\pi}{4}$ in standard position.

Describe his error.



Solution

Tyler incorrectly indicated the direction of the angle as positive.

or

Tyler sketched the reference angle, not $\theta = -\frac{7\pi}{4}$.

1 mark

Given the identity $\sec \theta + \cos \theta = \frac{2 - \sin^2 \theta}{\cos \theta}$,

- a) determine the non-permissible values of θ , over the interval $0 \leq \theta \leq 2\pi$.
- b) prove the identity for all permissible values of θ .

Solution

a) $\cos \theta = 0$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

1 mark for non-permissible values
($\frac{1}{2}$ mark for each value)

1 mark

Solution**Method 1**

b)

Left-Hand Side	Right-Hand Side
$\sec \theta + \cos \theta$	$\frac{2 - \sin^2 \theta}{\cos \theta}$
$\frac{1}{\cos \theta} + \cos \theta$	
$\frac{1 + \cos^2 \theta}{\cos \theta}$	1 mark for correct substitution of appropriate identities
$\frac{1 + 1 - \sin^2 \theta}{\cos \theta}$	1 mark for algebraic strategies
$\frac{2 - \sin^2 \theta}{\cos \theta}$	1 mark for logical process to prove the identity
	3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\sec \theta + \cos \theta$	$\frac{2 - \sin^2 \theta}{\cos \theta}$
	$\frac{2 - (1 - \cos^2 \theta)}{\cos \theta}$
	1 mark for correct substitution of appropriate identities
	$\frac{1 + \cos^2 \theta}{\cos \theta}$
	1 mark for algebraic strategies
	$\frac{1}{\cos \theta} + \frac{\cos^2 \theta}{\cos \theta}$
	1 mark for logical process to prove the identity
	$\sec \theta + \cos \theta$
	3 marks

Expand using the laws of logarithms.

$$\log\left(\frac{a}{b^4}\right)$$

Solution

$$\log a - \log b^4$$

1 mark for quotient law

$$\log a - 4\log b$$

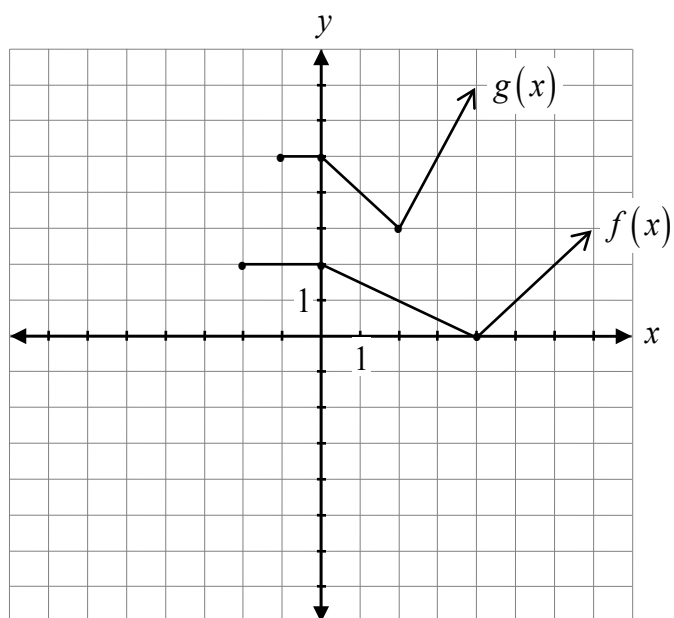
1 mark for power law

2 marks

Question 11

R4

State the equation of $g(x)$ in terms of $f(x)$.



Solution

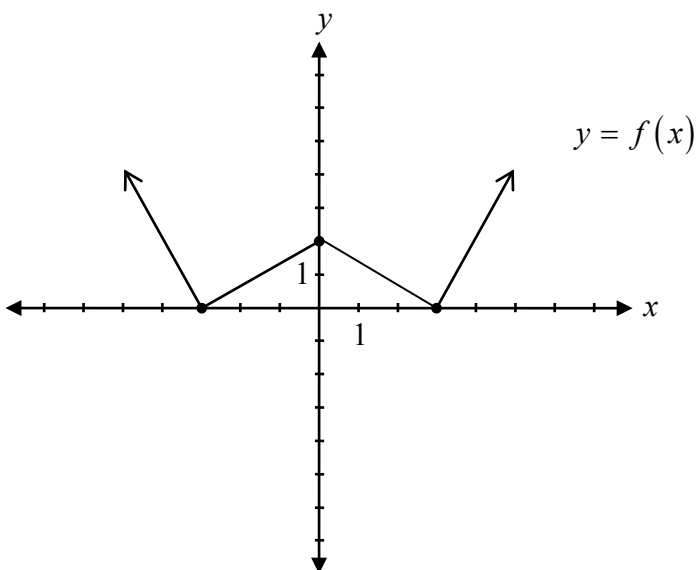
$$g(x) = f(2x) + 3$$

1 mark for horizontal compression

1 mark for vertical translation

2 marks

Explain why the inverse of the graph of $y = f(x)$ is not a function.

**Solution**

The domain of $f(x)$ was not restricted to ensure there is only one value of y for each x and one value of x for each y .

or

The graph of the inverse will not pass the vertical line test.

or

The graph of $f(x)$ does not pass the horizontal line test.

1 mark

Question 13

R10

Solve the following equation algebraically:

$$\log(x^2 + 5) - \log(x^2 + 1) = \log 3$$

Solution

$$\log\left(\frac{x^2 + 5}{x^2 + 1}\right) = \log 3$$

1 mark for quotient law

$$\frac{x^2 + 5}{x^2 + 1} = 3$$

½ mark for equating arguments

$$x^2 + 5 = 3(x^2 + 1)$$

$$x^2 + 5 = 3x^2 + 3$$

$$2 = 2x^2$$

$$1 = x^2$$

$$\pm 1 = x$$

½ mark for solving for x

2 marks

Describe the transformations used to obtain the graph of the function $y = 5f(x + 1)$ from the graph of $y = f(x)$.

Solution

Vertically stretch the graph by a factor of 5 and translate the graph one unit left.

1 mark for vertical stretch
1 mark for horizontal translation

2 marks

Question 15

P3

Solve algebraically:

$${}_nC_2 = 3n + 4$$

Solution

$$\frac{n!}{(n-2)!2!} = 3n + 4$$

½ mark for substitution into equation

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}} = 2!(3n + 4)$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n(n-1) = 2(3n + 4)$$

$$n^2 - n = 6n + 8$$

$$n^2 - 7n - 8 = 0$$

$$(n-8)(n+1) = 0$$

½ mark for rejecting the extraneous root

$$n = 8 \quad \cancel{n = -1}$$

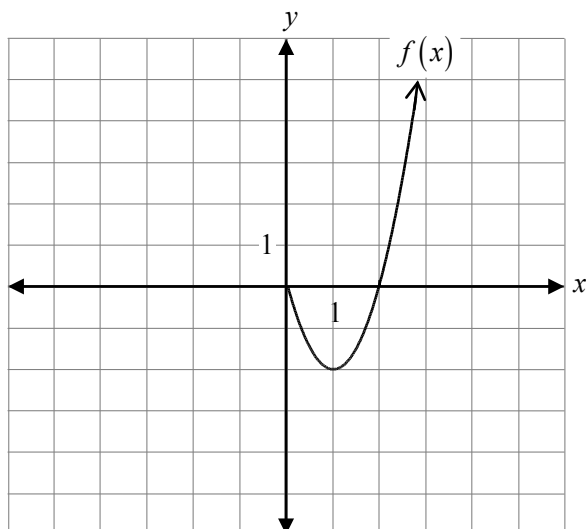
½ mark for the value of n

3 marks

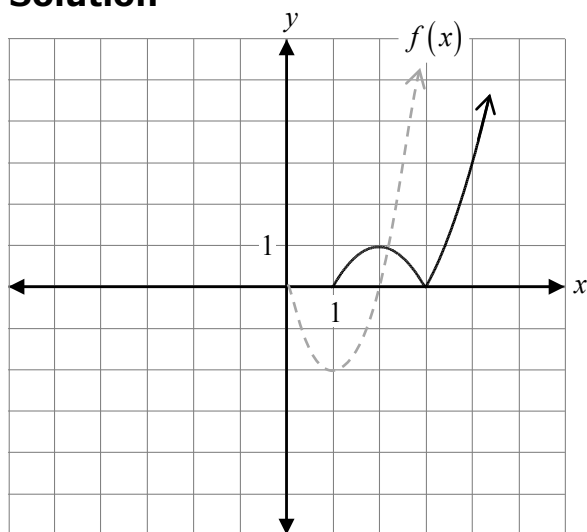
Question 16

R1, R2, R3

Given the graph of $f(x)$, sketch the graph of $y = \left| \frac{1}{2} f(x-1) \right|$.



Solution



1 mark for vertical stretch
1 mark for horizontal translation
1 mark for absolute value

3 marks

Explain why $f(x) = (x+2)^3(x-1)^{\frac{1}{2}}$ is not a polynomial function.

Solution

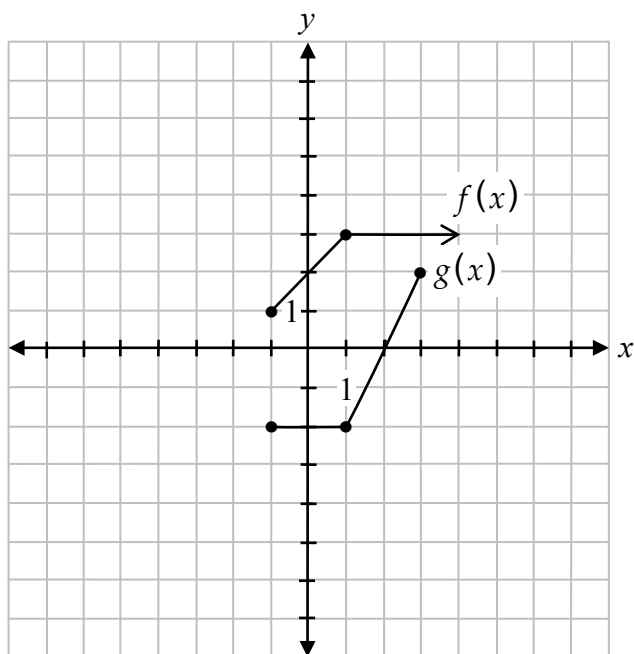
All factors in a polynomial function must have exponents that are whole numbers.

1 mark

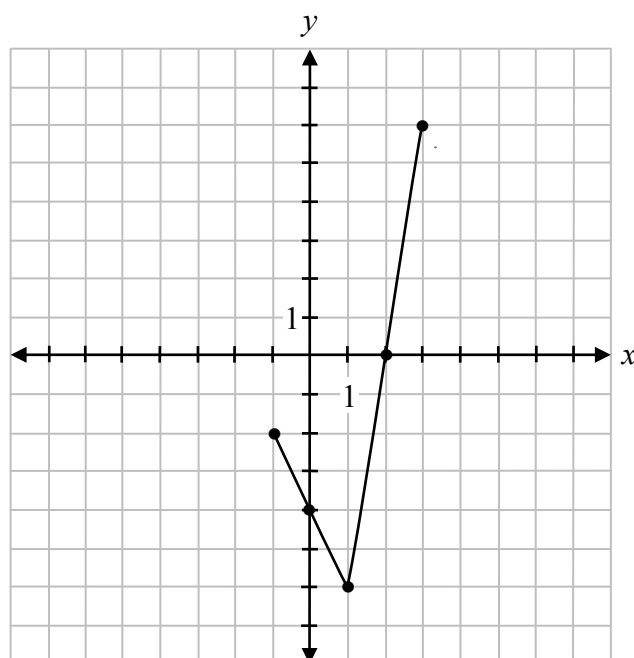
Question 18

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = (f \cdot g)(x)$.



Solution



x	$f(x)$	$g(x)$	$(f \cdot g)(x)$
-1	1	-2	-2
1	3	-2	-6
3	3	2	6

1 mark for operation of multiplication

1 mark for restricted domain

2 marks

Question 19

T6

Identify the trigonometric function that is equivalent to $\sin \frac{\pi}{4} \cos \frac{\pi}{3} + \cos \frac{\pi}{4} \sin \frac{\pi}{3}$.

a) $\sin \frac{2\pi}{7}$

b) $\sin \frac{7\pi}{12}$

c) $\cos \frac{2\pi}{7}$

d) $\cos \frac{7\pi}{12}$

Question 20

R7

Identify the logarithmic form of $5^x = 6$.

a) $\log_5 x = 6$

b) $\log_5 6 = x$

c) $\log_6 x = 5$

d) $\log_6 5 = x$

Question 21

T4

Given $f(\theta) = 3 \cos 2\theta - 1$ and $g(\theta) = \sin \theta + 1$, identify which statement is true.

- a) Both functions have the same period.
- b) Both functions have the same amplitude.
- c) Both functions have the same minimum value.
- d) Both functions have the same maximum value.

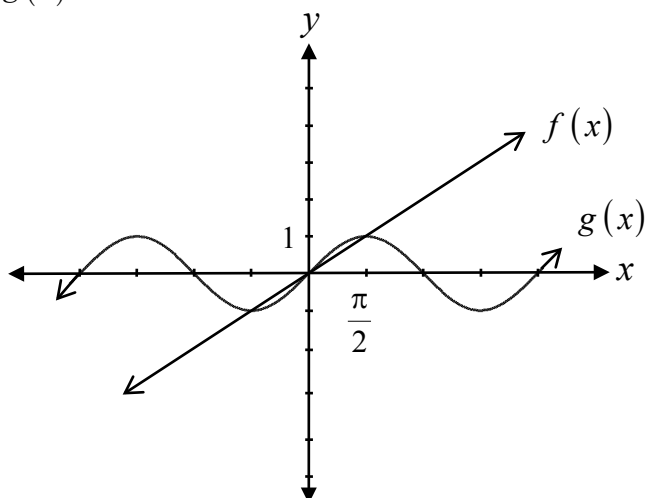
Question 22

P4

Identify the fourth term in the expansion of $(x + y)^5$.

- a) $10x^4y$
- b) $10x^3y^2$
- c) $10x^2y^3$
- d) $10xy^4$

Given the graphs of $f(x)$ and $g(x)$, identify the choice with all of the solutions of the equation $f(x) = g(x)$.



a) $x = -2\pi, -\pi, 0, \pi, 2\pi$

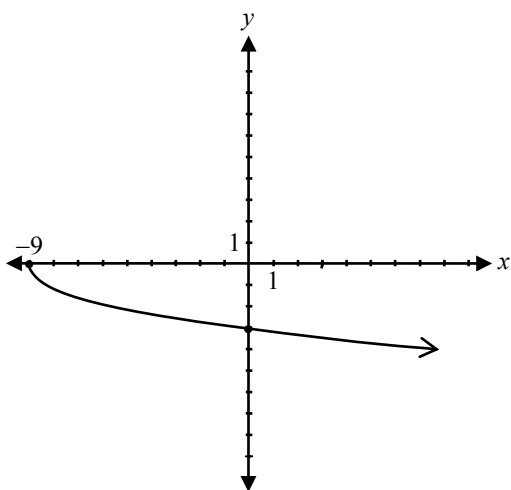
b) $x = -\frac{\pi}{2}, 0, \frac{\pi}{2}$

c) $x = \frac{\pi}{2}$

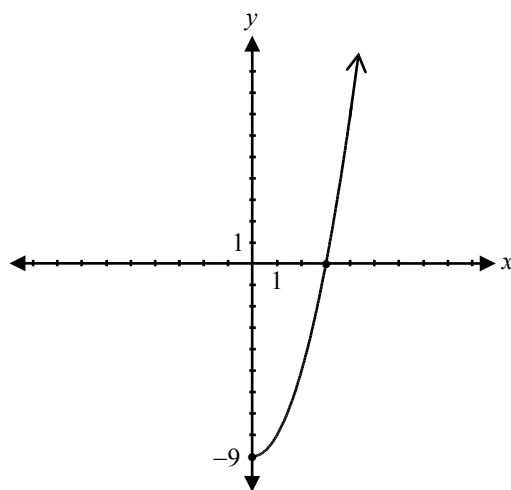
d) $x = -1, 0, 1$

Identify the graph of $f^{-1}(x)$ if $f(x) = x^2 - 9, x \geq 0$.

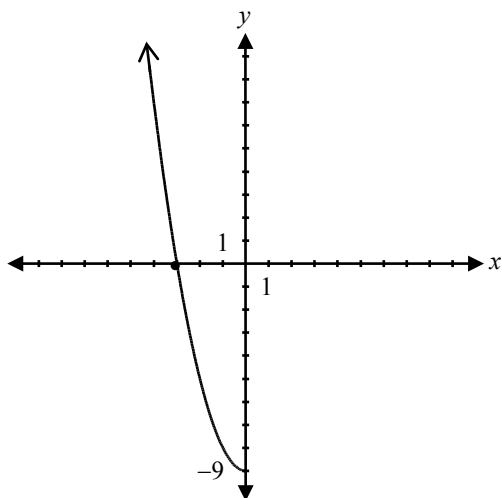
a)



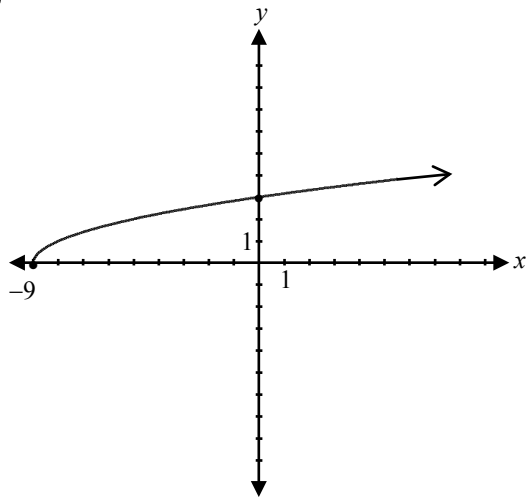
b)



c)



d)



Question 25

R11

Using the remainder theorem, identify which value of x results in a remainder of zero given

$$p(x) = x^3 + 7x^2 + 14x + 8.$$

- a) 1
- b) 0
- c) -1
- d) -3

Question 26

T3

Evaluate $\cos\left(\cos\left(\frac{3\pi}{2}\right)\right)$.

- a) 1
- b) $\frac{1}{2}$
- c) 0
- d) -1

Match the following equations with their graphs:

Solution

Place the appropriate letter
in this column.

$$f(x) = (x-1)^3(x+1)(x-3) \quad \underline{\hspace{2cm} C \hspace{2cm}}$$

$$g(x) = (x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm} B \hspace{2cm}}$$

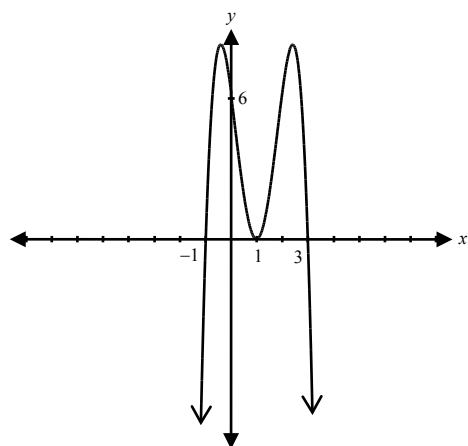
$$h(x) = -2(x-1)^2(x+1)(x-3) \quad \underline{\hspace{2cm} A \hspace{2cm}}$$

$$k(x) = 2(x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm} D \hspace{2cm}}$$

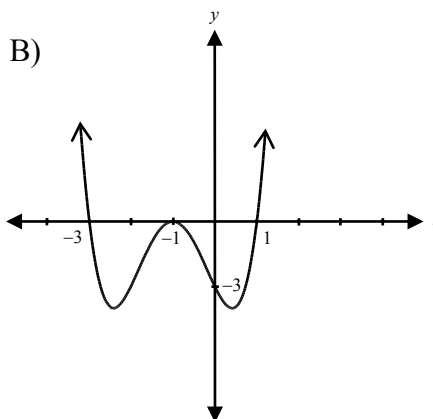
$\frac{1}{2}$ mark for each correct answer

2 marks

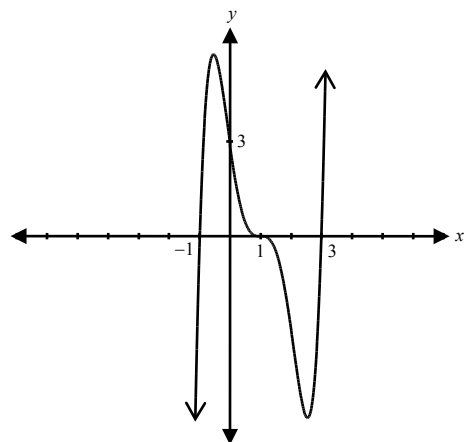
A)



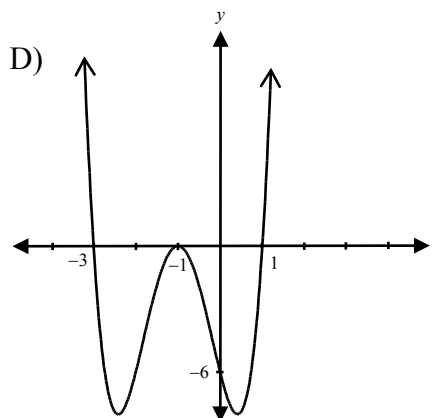
B)



C)



D)



If $\log 6 = p$, $\log 5 = r$ and $\log 2 = q$, express $\log 60$ in terms of p , q and r .

Solution

$$\log 60 = \log(6 \cdot 5 \cdot 2)$$

$\frac{1}{2}$ mark for combination

$$= \log 6 + \log 5 + \log 2$$

1 mark for product law

$$= p + r + q$$

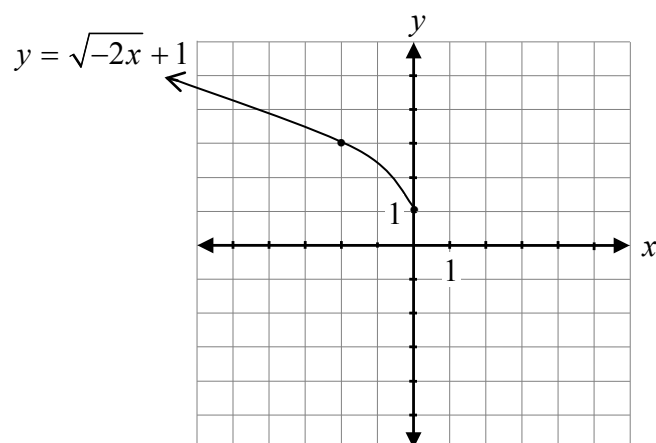
$\frac{1}{2}$ mark for substitution

2 marks

Sketch the graph of $y = \sqrt{-2x} + 1$.

Solution

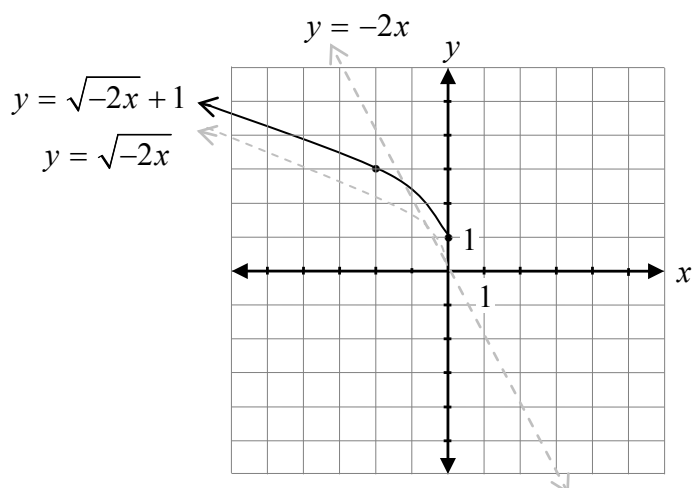
Method 1



- 1 mark for shape of a radical function
- 1 mark for horizontal compression
- 1 mark for vertical translation
- 1 mark for horizontal reflection

4 marks

Method 2



- ½ mark for shape between invariant points
- ½ mark for shape to the left of the invariant points
- 1 mark for invariant points where $y = 0$ and $y = 1$ (½ mark for each point)
- 1 mark for domain of $(-\infty, 0]$
- 1 mark for vertical translation

4 marks

Justify why the letters of the word FRANCE have a greater number of possible arrangements than the letters of the word CANADA.

Solution

CANADA has repeated letters that, when placed in different orders, do not change the arrangements of the letters.

$$\text{FRANCE: } 6! \qquad \text{CANADA: } \frac{6!}{3!}$$

1 mark

$\therefore$ France has a greater number of arrangements.

Question 31

R1, R14

Given $f(x) = \frac{1}{x-2}$ and $g(x) = x+5$,

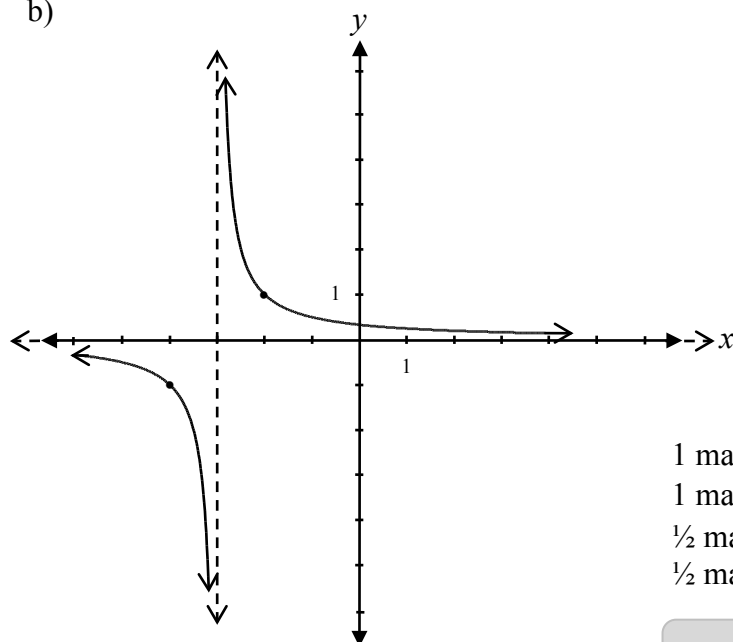
- determine the equation for $f(g(x))$.
- sketch the graph of $f(g(x))$.

Solution

$$\begin{aligned} \text{a) } f(g(x)) &= \frac{1}{(x+5)-2} \\ &= \frac{1}{x+3} \end{aligned}$$

1 mark

b)



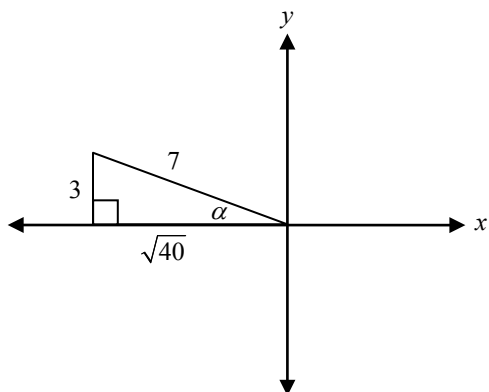
- 1 mark for asymptotic behaviour approaching $x = -3$
- 1 mark for asymptotic behaviour approaching $y = 0$
- $\frac{1}{2}$ mark for branch left of vertical asymptote
- $\frac{1}{2}$ mark for branch right of vertical asymptote

3 marks

Given that $\sin \alpha = \frac{3}{7}$, where α is in Quadrant II, and $\cos \beta = \frac{4}{5}$, where β is in Quadrant IV, determine the exact value of:

a) $\sin(\alpha - \beta)$

Solution

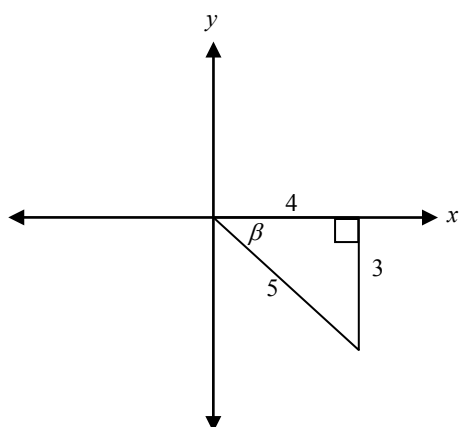


$$x^2 + y^2 = r^2$$

$$x^2 + 9 = 49$$

$$x^2 = 40$$

$$x = \pm\sqrt{40} \quad \frac{1}{2} \text{ mark for value of } x$$



$$x^2 + y^2 = r^2$$

$$16 + y^2 = 25$$

$$y^2 = 9$$

$$y = \pm 3 \quad \frac{1}{2} \text{ mark for value of } y$$

$$\begin{aligned} \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ &= \left(\frac{3}{7}\right)\left(\frac{4}{5}\right) - \left(\frac{-\sqrt{40}}{7}\right)\left(\frac{-3}{5}\right) \end{aligned}$$

$$= \frac{12}{35} - \frac{3\sqrt{40}}{35}$$

$$= \frac{12 - 3\sqrt{40}}{35} \quad \text{or} \quad \frac{12 - 6\sqrt{10}}{35}$$

$\frac{1}{2}$ mark for $\cos \alpha$

$\frac{1}{2}$ mark for $\sin \beta$

1 mark for substitution into correct identity

3 marks

Note(s):

- accept any of the following values for x : $x = \pm\sqrt{40}$, $x = -\sqrt{40}$ or $x = \sqrt{40}$
- accept any of the following values for y : $y = \pm 3$, $y = -3$ or $y = 3$

b) $\cos 2\alpha$ **Solution**

$$\begin{aligned}\cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= \left(\frac{-\sqrt{40}}{7}\right)^2 - \left(\frac{3}{7}\right)^2 \\ &= \frac{40}{49} - \frac{9}{49} \\ &= \frac{31}{49}\end{aligned}$$

or

$$\begin{aligned}\cos 2\alpha &= 2\cos^2 \alpha - 1 \\ &= 2\left(\frac{-\sqrt{40}}{7}\right)^2 - 1 \\ &= 2\left(\frac{40}{49}\right) - 1 \\ &= \frac{80}{49} - 1 \\ &= \frac{31}{49}\end{aligned}$$

or

$$\begin{aligned}\cos 2\alpha &= 1 - 2\sin^2 \alpha \\ &= 1 - 2\left(\frac{3}{7}\right)^2 \\ &= 1 - 2\left(\frac{9}{49}\right) \\ &= 1 - \frac{18}{49} \\ &= \frac{31}{49}\end{aligned}$$

1 mark for substitution of an appropriate identity

1 mark

Determine the domain and range of $f(x) = \sqrt{x-5} - 1$.

Solution

Domain: $\{x \in \mathbb{R} \mid x \geq 5\}$ **or** $[5, \infty)$ 1 mark for domain

Range: $\{y \in \mathbb{R} \mid y \geq -1\}$ **or** $[-1, \infty)$ 1 mark for range

2 marks

Justify why 4.7 is a better estimate than 4.3 for the value of $\log_2 26$.

Solution

$$2^4 = 16 \quad 2^5 = 32$$

or

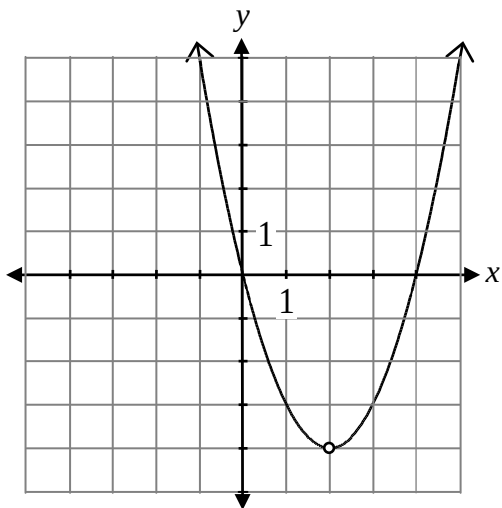
$$\log_2 16 = 4 \quad \log_2 32 = 5$$

26 is closer to 32 than 16; therefore $\log_2 26$ is closer to 5 than 4.

1 mark

Sketch the graph of the function:

$$f(x) = \frac{x(x-2)(x-4)}{(x-2)}$$

Solution

1 mark for point of discontinuity (hole) at $(2, -4)$
($\frac{1}{2}$ mark for x value, $\frac{1}{2}$ mark for y value)

$\frac{1}{2}$ mark for shape of a parabola

$\frac{1}{2}$ mark for end behaviour

2 marks

Evaluate:

$$\sec^2\left(\frac{\pi}{6}\right) + \tan\left(\frac{7\pi}{6}\right) \csc\left(-\frac{2\pi}{3}\right)$$

Solution

$$\left(\frac{2}{\sqrt{3}}\right)^2 + \left(\frac{\sqrt{3}}{3}\right)\left(-\frac{2}{\sqrt{3}}\right)$$

1 mark for $\sec\left(\frac{\pi}{6}\right)$ (½ mark for value, ½ mark for quadrant)

$$\left(\frac{4}{3}\right) - \left(\frac{2}{3}\right)$$

1 mark for $\tan\left(\frac{7\pi}{6}\right)$ (½ mark for value, ½ mark for quadrant)

$$\frac{2}{3}$$

1 mark for $\csc\left(-\frac{2\pi}{3}\right)$ (½ mark for value, ½ mark quadrant)

3 marks

The graph of $f(x) = 3x + 7$ is reflected over the y -axis.

Determine the equation of the new function.

Solution

$$y = -3x + 7$$

or

$$y = f(-x)$$

1 mark

Determine the equations of all of the asymptotes of the function:

$$y = \frac{2x + 1}{x - 3}$$

Solution

Horizontal asymptote at $y = 2$

1 mark for horizontal asymptote

Vertical asymptote at $x = 3$

1 mark for vertical asymptote

2 marks

Question 39

R11

One of the zeros of $p(x) = x^3 + 6x^2 - 32$ is $x = 2$. Determine all of the other zeros of $p(x)$.

Solution

2		1	6	0	-32
		↓	2	16	32
		1	8	16	0

1 mark for synthetic division (or any equivalent strategy)

$$0 = x^2 + 8x + 16$$

½ mark for the other factors

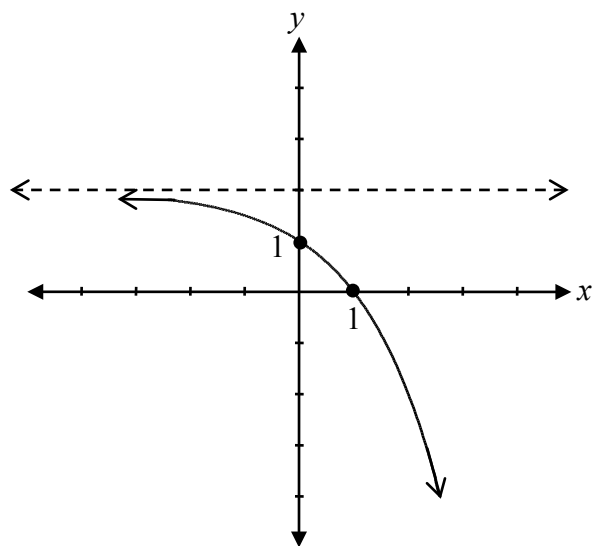
$$0 = (x + 4)(x + 4)$$

$$x = -4$$

½ mark for the other zeros

2 marks

Sketch the graph of $y = -2^x + 2$.

Solution

1 mark for shape of an exponential function

1 mark for vertical reflection

1 mark for asymptotic behaviour approaching $y = 2$

3 marks

Given the function $f(x) = \frac{2}{x} - 1$, justify why $f(f(2))$ is undefined.

Solution

$$f(2) = \frac{2}{2} - 1$$

$$= 1 - 1$$

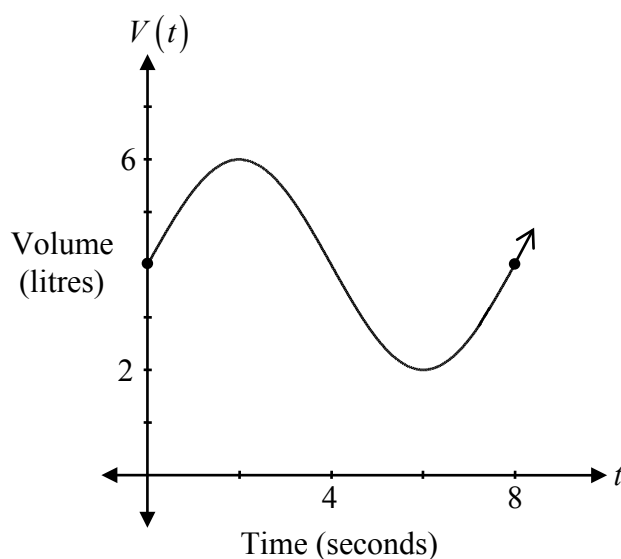
$$= 0$$

$f(f(2)) = \frac{2}{0} - 1$, which is undefined because the denominator cannot be zero.

1 mark for justification

1 mark

The following graph represents the volume of air in an adult's lungs. If $V(t)$ is the volume of air in litres and t is the time in seconds, determine an equation that represents this sinusoidal function.

**Solution**

$$V(t) = 2 \sin\left(\frac{\pi}{4}t\right) + 4$$

or

$$V(t) = 2 \cos\left[\frac{\pi}{4}(t - 2)\right] + 4$$

1 mark for amplitude

$\frac{1}{2}$ mark for period

$\frac{1}{2}$ mark for consistent value of b

1 mark for vertical translation

3 marks

Explain why the domain of $y = \log_2(x - 1)$ is $x > 1$.

Solution

The argument of a logarithmic function must be positive.

1 mark

The point $(-2, 7)$ is on the terminal arm of an angle in standard position.

Determine the coordinates of the corresponding point, $P(\theta)$, on the unit circle.

Solution

$$x^2 + y^2 = r^2$$

$$(-2)^2 + (7)^2 = r^2$$

½ mark for substitution of $x = \pm 2$ and $y = 7$

$$4 + 49 = r^2$$

$$53 = r^2$$

½ mark for solving for r

$$\sqrt{53} = r$$

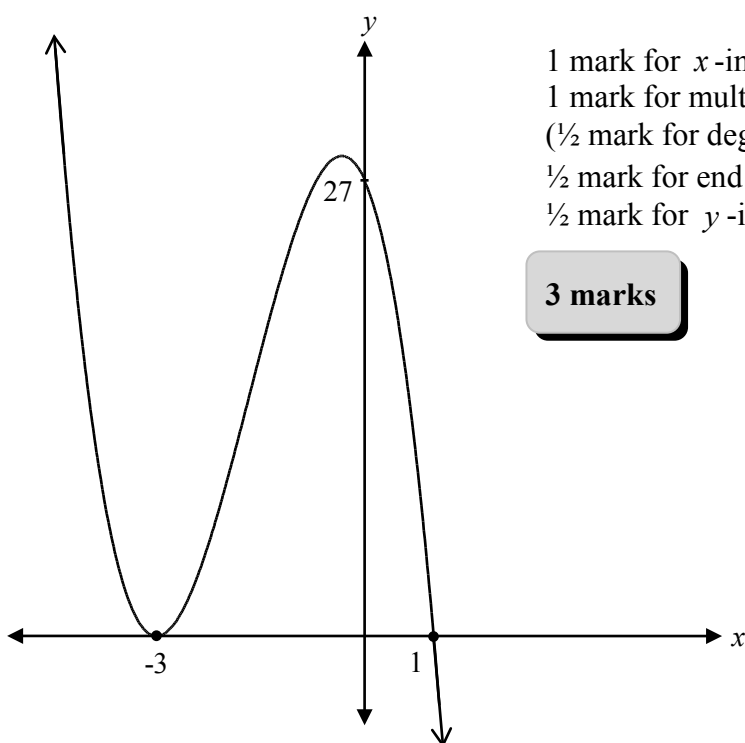
1 mark for $P(\theta)$ (½ mark for each coordinate)

$$P(\theta) = \left(\frac{-2}{\sqrt{53}}, \frac{7}{\sqrt{53}} \right)$$

2 marks

Sketch a graph of $P(x)$ that satisfies all of the following conditions:

- $P(x)$ is a polynomial function of degree 3.
- $P(x)$ has a zero at -3 with a multiplicity of 2.
- $P(x)$ has a zero at 1.
- $P(x)$ has a leading coefficient of -3 .

Solution

1 mark for x -intercepts
1 mark for multiplicity
($\frac{1}{2}$ mark for degree 2 at $x = -3$, $\frac{1}{2}$ mark for degree 1 at $x = 1$)
 $\frac{1}{2}$ mark for end behaviour
 $\frac{1}{2}$ mark for y -intercept

3 marks

Question 46

P1

Determine how many 4-digit numbers greater than 4000 can be made using the digits 2, 3, 4, 5, and 6 if repetitions are not allowed.

Solution

$$\frac{3}{4, 5, 6} \cdot \frac{4}{\text{remaining options}} \cdot \frac{3}{\text{remaining options}} \cdot \frac{2}{\text{remaining options}} = 72$$

1 mark for restriction of first digit

1 mark for fundamental counting principle

2 marks